

APPLIED STATISTICS, PROBABILITY THEORY AND MATHEMATICAL STATISTICS

Attila Barta, Sándor Pecsora

Combinatorics

Description to this presentation

After solving each exercise with „pen and paper”, also solve them with Matlab as well.

If the **title of the exercise** is red, skip on the laboratory class, that is part of the lecture slide!

Permutations

The number of permutation without repetition is:

$$P_n = n! = n \cdot (n - 1) \cdot \dots \cdot 2 \cdot 1$$

The number of permutation with repetition is:

$$P_n^{r_1, r_2, \dots, r_s} = \frac{n!}{r_1! \cdot r_2! \cdot \dots \cdot r_s!}$$

Where r_i is the number of repetitions of element i .

Permutations in Matlab

All possible Permutations of 3 distinguishable elements:

```
perms([1 2 3])
```

ans =

3	2	1
3	1	2
2	3	1
2	1	3
1	3	2
1	2	3

The number of possible Permutations of 3 distinguishable elements:

```
factorial(3)
```

ans =

6

Permutations in Matlab

All possible Permutations with repetition:

```
unique(perms([1 1 3]), 'rows')
```

ans =

1	1	3
1	3	1
3	1	1

All possible Permutations with repetition:

```
unique(perms([repelem(2,3) repelem(1,2) repelem(3,1)]), 'rows')
```

Ordered selections

The number of ordered selections without repetition is:

$$V_n^k = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot (n - k + 1) = \frac{n!}{(n - k)!}$$

The number of ordered selections with repetition is:

$$V_n^{k,r} = \underbrace{n \cdot n \cdot \dots \cdot n}_k = n^k$$

Ordered selections in Matlab

The number of ordered selections without repetition of 4 elements out of 8 is:

```
prod(5:8)
```

```
ans=
```

```
1680
```

Since:

$$V_n^k = \frac{n!}{(n-k)!} = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1)$$

$$V_8^4 = \frac{8!}{(8-4)!} = 8 \cdot 7 \cdot 6 \cdot 5 = 1680$$

Ordered selections in Matlab

The number of ordered selections with repetition of 4 elements out of 8 is:

With caret symbol (ascii code: alt+94=^):

```
8^4
```

```
ans=
```

```
4096
```

```
power(8,4)
```

```
ans=
```

```
4096
```


Combinations

The number of combinations without replacement is:

$$C_n^k = \binom{n}{k} = \frac{n!}{k! \cdot (n - k)!}$$

The number of combinations with replacement is:

$$C_n^{k,r} = \binom{n + k - 1}{k} = \frac{(n + k - 1)!}{k! \cdot (n - 1)!}$$

Combinations in Matlab

Number of combinations

```
b = nchoosek(5,4)
```

```
b =
```

5

Number of combinations

```
v = 2:2:10;
```

```
C = nchoosek(v,4)
```

```
C = 5×4
```

2	4	6	8
2	4	6	10
2	4	8	10
2	6	8	10
4	6	8	10

Combinations in Matlab

Number of combinations

```
v = uint16([10 20 30]);  
C = nchoosek(v,uint16(2))
```

C = 3x2 uint16 matrix

10	20
10	30
20	30

Number of combinations

```
C = combnk(1:4,2)
```

C = 6x2

3	4
2	4
2	3
1	4
1	3
1	2

Exercise 1.2

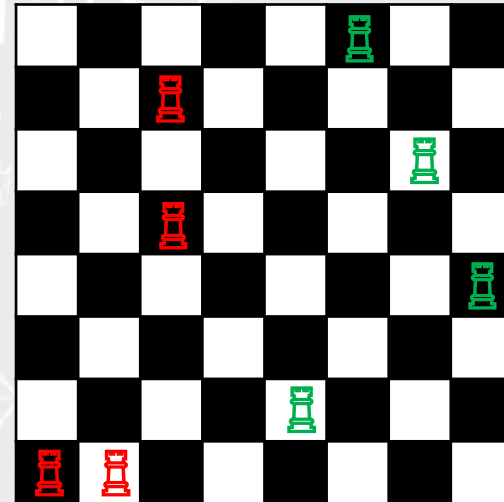
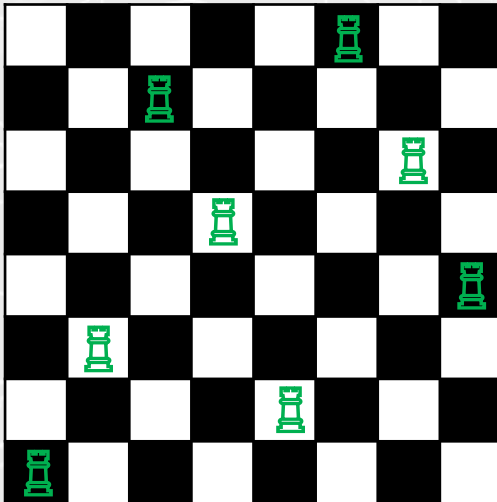
Find the number of possible arrangements of 8 castles on the chess board in a way that they do not hit each other?

What is the result if we can distinguish between the castles?

$$P_8 = 8!$$

$$P_8 \cdot P_8$$

Help:



Exercise 1.3

How many real four digit numbers (they can not start with zero) can be formed from digits 0, 1, 2, 3, 4, 5, 6?

$$V_6^1 \cdot V_6^3$$

Exercise 1.4

We have 12 books on the shelf. How many ways can the books be arranged on the shelf if 3 particular books must to be next to each other

- a) if the order of the three books does not count?
- b) if the order of the three books does count?

a) $P_{10} = 10!$

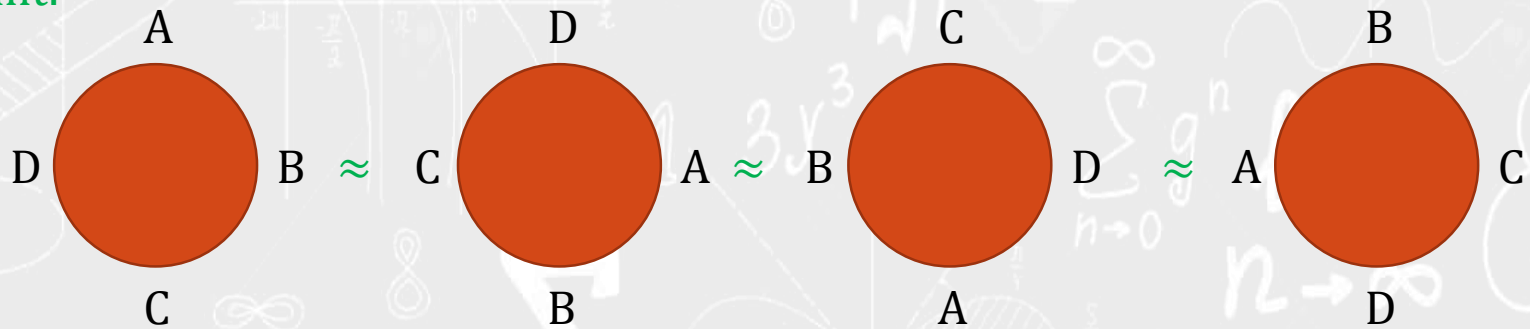
b) $P_{10} \cdot P_3 = 10! \cdot 3!$

Exercise 1.6

$$P_6 = 6!$$

In how many ways can 7 people be arranged around a round table?

Hint:



Exercise 1.7

In how many ways can 5 men and 5 women be arranged around a round table if neither two men, nor two women can sit next to each other?

$$P_4 \cdot P_5 = 4! \cdot 5!$$

Exercise 1.9

Three postmen must deliver six letters. Find the number of possible distribution of the letters. $V_3^{6,r} = 3^6$

Exercise 1.11

Find the number of possible fillings of a lottery coupon (5 numbers from 90).

$$C_{90}^5 = \frac{90!}{5! \cdot (90 - 5)!}$$

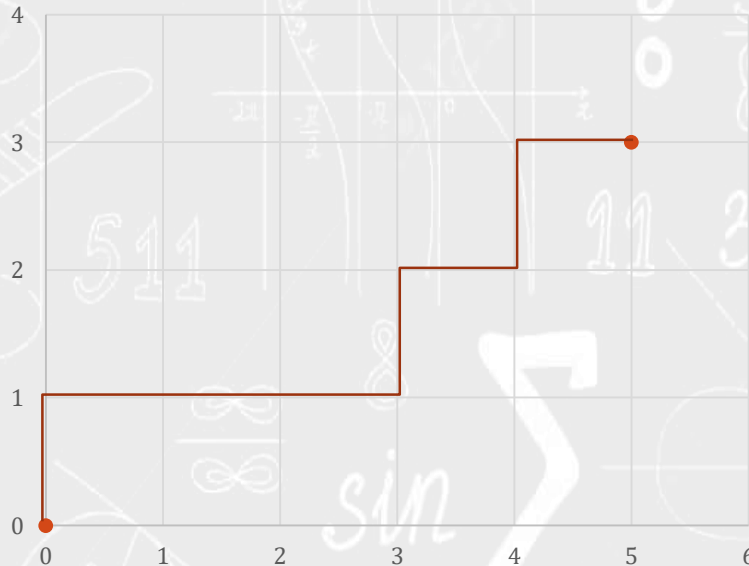
Exercise 1.12

Find the number of possible paths from the origin to the point (5; 3) if we can walk only on points with integer coordinates and we can step only upwards and right.

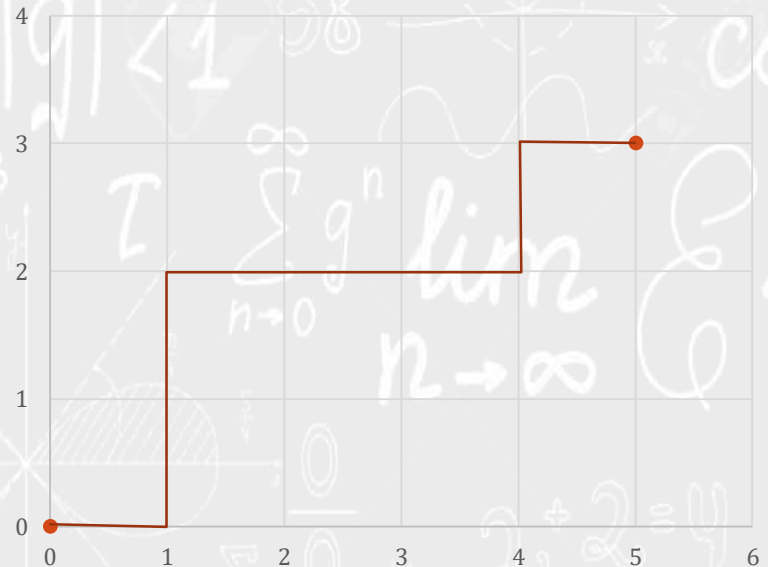
$$p_8^{3,5}$$

Hint:

Example path 1



Example path 2



Exercise 1.20

A deck of ordinary cards is shuffled, and 10 cards are dealt.

- a. In how many cases will we have aces among the 10 cards?
- b. Exactly one ace?
- c. At most one ace?
- d. Exactly two aces?
- e. At least two aces?

$$C_{52}^{10} - C_{48}^{10}$$

$$C_{48}^9 \cdot C_4^1$$

$$C_{48}^9 \cdot C_4^1 + C_{48}^{10} \cdot C_4^0$$

$$C_{48}^8 \cdot C_4^2$$

$$C_{48}^8 \cdot C_4^2 + C_{48}^7 \cdot C_4^3 + C_{48}^6 \cdot C_4^4$$

Or

$$C_{52}^{10} - (C_{48}^9 \cdot C_4^1 + C_{48}^{10} \cdot C_4^0)$$

Exercise 1.22

Find the number of real six digit numbers having three odd and three even digits.

Starts with an even number: $4 \cdot 5^5$
Starts with an odd number: 5^6
Overall: $C_5^2(4 \cdot 5^5 + 5^6)$

Hint:

First digit is odd

1,3,5,7,9

First digit is even

0,2,4,6,8

Exercise 1.25

In how many ways we can distribute 7 apples and 9 peaches among 4 kids?

$$C_4^{7,r} \cdot C_4^{9,r}$$

Literature

- ❑ Ágnes Baran: Mathematics for engineers 1. (Laboratory slides)
- ❑ Sándor Baran: Probability theory and statistics
- ❑ Fazekas István: Valószínűségszámítás, Debreceni Egyetemi Kiadó, 2009
- ❑ Fazekas István: Bevezetés a matematikai statisztikába, Debreceni Egyetemi Kiadó, 2009
- ❑ Matlab examples:
<https://www.mathworks.com/help/examples.html>